From Design to Evidence: Implementing and Auditing a Risk-Triage Pipeline for Kenyan Public Procurement

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***Abstract*—This work implements the preprocessing and modeling pipeline designed in Phase 1 for ranking 108,984 awarded Kenyan public contracts by integrity risk and tests each design decision against the real data. Several Phase 1 assumptions did not hold. Missingness is strongly associated with the recording year rather than the procurement method. The supplier identifier does not behave as a stable company identifier. About half of the merges made at a fuzzy-matching threshold of 90 join different firms. Upper-tail winsorization alone does not reduce skew after the log transform, and the composite score depends on which indicators are available. Each finding led to a recorded, evidence-based decision. All cleaning, imputation, encoding, and scaling run inside a scikit-learn pipeline fitted on the 2018 to 2023 training period only. The pipeline maps $72{,}585 \times 23$ raw records to $72{,}585 \times 14$ model inputs without losing rows. Supplier entity resolution reaches precision of 0.98 and recall of 0.90 on a held-out labeled sample. The final model flags 1,131 of 36,399 held-out contracts (3.1%) above a threshold fixed at the training top 5%. Synthetic tests show much stronger sensitivity to early signing (recall 0.842) than to value inflation (recall 0.117).**

I. Introduction

Public procurement is a major channel for corruption, but verified corruption labels do not exist for Kenyan contract data. Phase 1 therefore framed the task as unsupervised risk ranking. A composite red-flag score $R(c)$, following the corruption risk index of Fazekas et al. [1], is combined with an Isolation Forest [2] to produce a review queue for auditors.

Phase 1 was a design based on a first look at the data. This report audits it. For each planned step, the report states what was planned, what the data showed, and what was changed, together with the statistic or figure that justifies the decision. Section XI summarizes these decisions in a side-by-side log (Table VI).

Three validation exercises go beyond the original plan. Supplier entity resolution is validated on hand-labeled pairs, with a pre-specified held-out test. The single-bidding indicator is checked against the public procurement portal. The model is tuned with synthetic anomaly injection. Code, tests and figures are available in the project repository[1].

II. Data and setup

The data is the Kenyan Public Procurement Regulatory Authority release in the Open Contracting Data Standard [3]. Merging the release gives a table of $277{,}325 \times 23$. The scoring unit is the awarded contract: 108,984 contracts with a tender closing date between July 2018 and August 2026. Contracts are split by tender closing date, so the model always learns from the past. The training period (2018 to 2023) has 72,585 contracts, and the test period (2024 to August 2026) has 36,399. The final year is therefore partial.

$R(c)$ is the mean of the available indicators. These are f_1 (direct award), f_3 (short decision period, an empirical CDF score of days from tender close to signing), f_4 (the share of the buyer's earlier spending won by the same supplier), and f_5 (Benford deviation [4] of the buyer's earlier award values). Buyer-level indicators use expanding windows. A contract sees only awards dated strictly before it, so no future information reaches any feature.

III. Missingness

Fig. 1 shows the missingness pattern of the training set. The most incomplete fields are the award criteria (73.4%), the contract start date (50.9%), the supplier name (33.8%), the contract end date (11.2%) and the signing date (6.4%). Little's test [5] on the numeric block (log value, days to signing and the two contract-period offsets) gave $\chi^2 = 2{,}526.8$ ($\mathrm{df} = 16$, $p \approx 0$, eight patterns), so the data is not missing completely at random.

Phase 1 expected missingness to depend on the procurement method. It does not. Cramér's V between missingness and competition level was 0.024 to 0.074. Against the recording year, it was 0.101 to 0.903. The evidence rejects MCAR and shows that missingness is strongly associated with recording year. For preprocessing, the missingness is treated as consistent with MAR conditional on the recording period, while acknowledging that MNAR cannot be ruled out. The strongest case is the award criteria, which were 0.3% missing in 2018 and 98.5 to 100% missing from 2022 (V = 0.903). Publication of the field effectively ceased from 2022 in this release, so its absence is not a red flag. Indicator f_2 was removed from $R(c)$ and from the model.

Year itself cannot serve as an imputation group, because the test years never occur in training. Imputation therefore uses groups that exist in both periods. Value is imputed with the median of its procurement category, days to signing with the

[1]https://github.com/Razan-O-Elobeid/kenya-procurement-triage

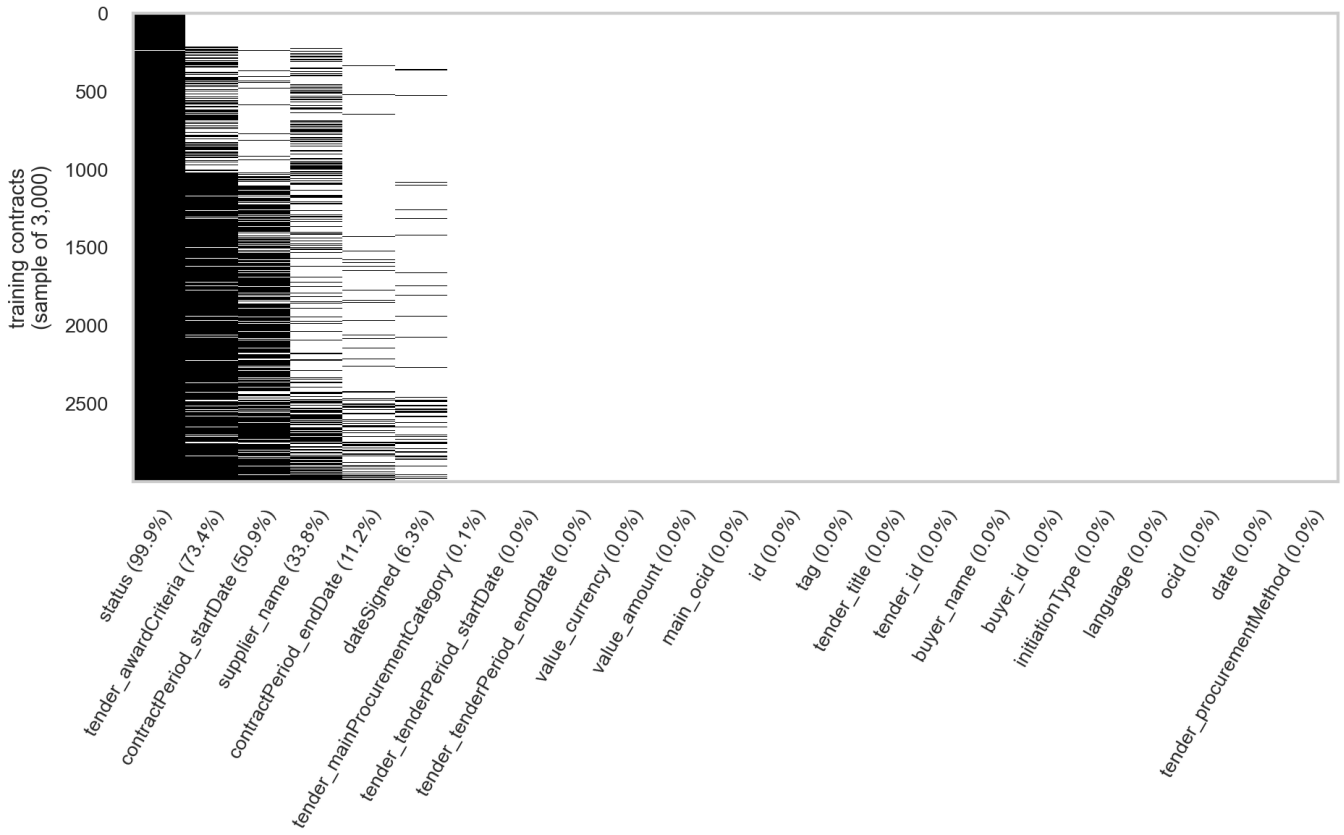


Fig. 1. Missingness pattern of the training set (sample of 3,000 contracts, sorted by number of missing fields; black is missing).

median of its procurement method, and the category with the mode of its method. f_4 and f_5 use the training median plus a missing-value indicator, so the model can still see that a value was absent.

IV. Indicators and the Risk Score

A. Benford coverage

Phase 1 required at least 100 earlier awards per buyer before computing f_5. On training data this gave a coverage of only 62.6%, which excludes small buyers. A buyer-group fallback with a minimum of 10 awards raised coverage to 92.9%.

B. Pre-closure signing

A binary flag `is_signed_pre_closure` records contracts signed before bidding closed. This keeps the compliance signal explicit instead of folding it into a continuous score. It affects 5.4% of training contracts and 9.0% of test contracts. Fig. 2 shows the distribution of $R(c)$ in both periods.

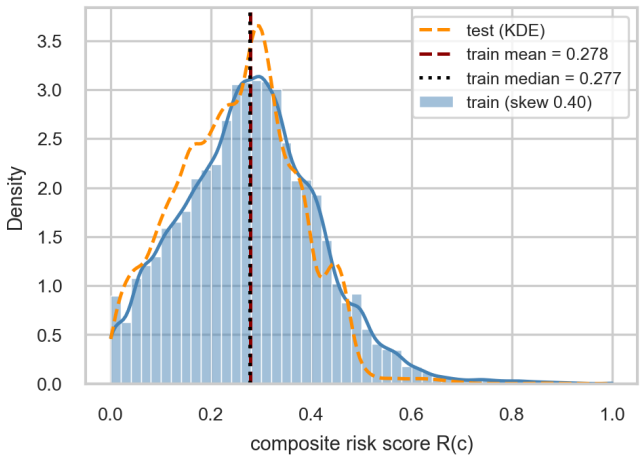


Fig. 2. Distribution of the composite risk score $R(c)$, training histogram with test density.

C. Supplier entity resolution

Supplier names appear in many forms, which splits one firm into several and weakens f_4. Phase 1 planned three stages: name cleaning, fuzzy matching at a token sort ratio of 90 or more within first-token blocks, and a veto using `identifier_id`.

The veto proved impossible. The release holds 552,087 supplier-party rows with 552,087 distinct identifiers under the scheme “Bidders”. In this export, `identifier_id` behaves as a bid or party-record identifier rather than a stable company identifier. This is consistent with the registry’s note that the

supplier identifier scheme is invalid [3]. Blocking was also dropped because comparing all 21,963 training names took 4 seconds.

A stratified development set of 200 candidate pairs was manually labeled without seeing similarity scores; 184 were decidable. Candidate pairs were labeled using the supplier names together with targeted internet searches, including company websites and other publicly available sources where available. Pairs were marked undecidable when the available evidence was insufficient to determine whether the names referred to the same firm. At threshold 90, precision was only 0.54 (95% bootstrap interval 0.42 to 0.67). Shared generic words such as "enterprises" and "supplies" inflate similarity between unrelated firms. Precision was 0.80 for high-volume suppliers but 0.48 for small ones. An acronym rule reached precision 0.35 and was rejected. A strict threshold of 95 (Option B) gave precision 0.92 but recall 0.39.

The development set showed that most true duplicates differ only by legal forms, connectors and plurals. Option A removes LTD, CO, AND, THE, M/S, T/A, and plural S, merges identical cores, and fuzzy-matches cores at 95 or more. Because Option A was designed after inspecting the development set, its rule and an adoption criterion were written down before a fresh held-out test sample of 60 pairs was drawn and labeled (59 decidable). The criterion was precision of at least 0.90 and recall above Option B on the same pairs.

Table I reports the result. Option A reached precision 0.981 and recall 0.895, against 0.702 and 0.246 for Option B on the same pairs. All 15 identical-core pairs and 14 of 15 fuzzy pairs were correct. Exact binomial lower bounds [6] for these groups are 0.78 and 0.68, wider than the bootstrap suggests. The held-out sample shows a large improvement for Option A relative to the uncertainty observed. With 59 labeled pairs, however, the estimates remain imprecise, so this is strong evidence on the held-out sample rather than proof of population-level performance. A fairness check supports the comparison: only 2 of the 1,434 pairs that Option B would merge lie outside Option A's candidate set.

TABLE I
ENTITY RESOLUTION ON THE HELD-OUT TEST SAMPLE (59 LABELED PAIRS, WEIGHTED TO STRATUM SIZES, 95% BOOTSTRAP INTERVALS; FMR IS THE FALSE-MATCH RATE)

Rule	Precision	Recall	FMR
A: core ≥ 95	0.981 (0.94, 1.00)	0.895 (0.77, 1.00)	0.003
B: original ≥ 95	0.702 (0.44, 1.00)	0.246 (0.13, 0.40)	0.019
Sensitivity checks, not used to choose			
A: core ≥ 90	0.608	1.000	0.120
A: identical cores only	1.000	0.647	0.000

The resolver builds supplier clusters from training names only. Test-period names are matched against this frozen reference with the same rule, and ties go to the more frequent training identity. Names that cannot be linked receive new identities. The 21,963 training names became 19,247 identities, and the largest cluster holds 13 name variants of one firm. Linking pairs transitively can join names that never matched directly, so a cluster check was fixed before running. The eight largest clusters and the one chained cluster with a direct score below 90 were inspected, and all were single firms. Of 17,107 test-period names, 4,558 occur in training, 1,948 were linked to a training identity, and 10,601 could not be linked to a training-period identity and were assigned new identities. Resolution changed f_4 by more than 0.01 for 1.2% of training and 1.8% of test contracts. The resolver passes 23 synthetic tests, including leakage tests.

D. Single bidding

Phase 1 left f_6 (exactly one bidder) pending validation. The supplier-party records carry only the roles "supplier" and "buyer". A peak at three parties per process, with few processes at two, is consistent with the statutory minimum for requests for quotations, which should ordinarily go to at least three persons [7]. Some party records may therefore reflect invitations or quotation participation rather than submitted bids.

The acceptance rule was written before the check. After a three-process pilot showed that the portal sometimes reports "number of bids received", a minimum of 30 verified processes was added. This minimum was a pre-specified judgment call, not a formal power calculation. Bid counts could be verified for 15 of 50 sampled processes (30%, exact interval 18 to 45%), and the portal's search was slow and inconsistent. Every independently verifiable sampled process agreed with the data (15 of 15, exact interval 78 to 100%). Even so, 15 perfect agreements cannot establish a true agreement rate above 90%: the exact lower bound is 0.78 at 15, 0.88 at 30 and 0.90 at 35. Only two verified processes had three or more parties, so the invitation question remains open. All 50 sampled processes were open tenders, so the check does not establish what party counts mean for selective procedures. f_6 stays outside $R(c)$, pending validation. On training data, 18.6% of competitive contracts with recorded supplier parties have exactly one, the candidate f_6 condition.

V. OUTLIERS

Fig. 3 shows the outlier structure of the training data. The IQR rule (Q1 = 110,106 KES, Q3 = 2,884,340 KES) puts the upper bound for contract value at 7.05M KES, with 12.7% of awards above it. These are concentrated in works (21.2%) and services (11.5%) rather than goods (5.5%). Capping at the IQR bound would therefore flatten legitimate large contracts, which confirms the Phase 1 choice of IQR for detection only.

Phase 1 planned to winsorize at the training P99 and then apply $\log(1+x)$. The evidence changed this plan. With zeros excluded, log value has skew -0.62, and an upper cap alone makes it -0.72. The remaining skew comes from the lower tail. There are 663 awards under 100 KES, values implausibly small for the observed procurement context and potentially reflecting unit or entry errors. Clipping at both the training P1 (212 KES) and P99 (164.9M KES) gives skew -0.24 and kurtosis of 0.54. The capped distribution is shown in the second

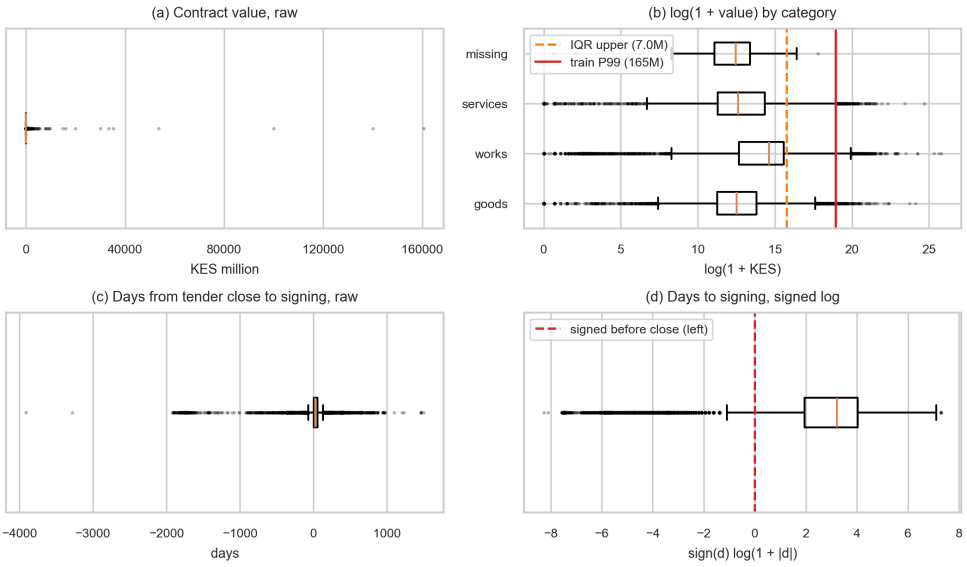


Fig. 3. Outliers in contract value and decision period (training data). (a) raw value; (b) $\log(1 + \text{value})$ by category with the IQR bound and the training P99; (c) raw days to signing; (d) signed log of days.

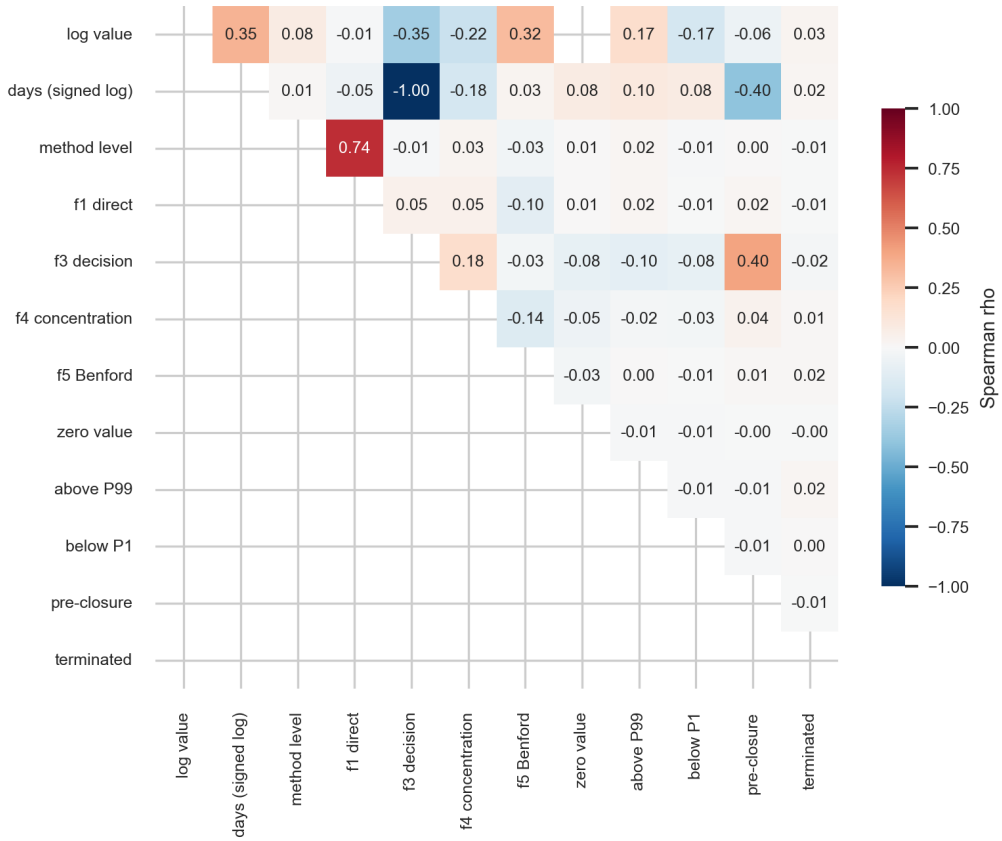


Fig. 4. Spearman correlation between candidate features (training data). Log value and zero value never co-occur, so their cell is empty.

column of Fig. 7, which summarizes the full transformation sequence. Zeros are treated as missing for the continuous transformation and kept as the `zero_value_award` flag. A new flag `below_p1` ($0 < \text{value} < P1$) mirrors `above_p99`, so clipping erases neither signal.

Days to signing range from $-3{,}911$ to 1,499, with 5.4% neg-

ative. Signings more than a year before close are concentrated in 2022 and 2023 (419 of 444). This is a systematic pattern whose cause the data cannot identify. The variable is clipped at the training P1 and P99 (-310 and 321 days) before the signed log $\operatorname{sign}(d)\log(1+|d|)$. The clipping bounds the range for the Isolation Forest, whose splits are drawn between each feature's minimum and maximum. It is not a skew-reduction step. All 3,905 negative values stay negative, and the pre-closure flag keeps them explicit.

VI. Multicollinearity

The Phase 1 protocol screens pairs with Spearman $|\rho| \geq 0.85$, then checks VIF above 10 on standardized inputs. Association with $R(c)$ is never used to choose between collinear features, because that would be circular. Fig. 4 shows the correlations after the outlier transforms. Only one pair crosses the threshold: days to signing and f_3 ($\rho = -1.00$), which Phase 1 expected by construction (Fig. 5a). Method level and f_1 ($\rho = 0.74$, VIF 5.3) fall below both thresholds (Fig. 5b). They are still separated, because f_1 is a deterministic function of the method. The empirical screen alone would not have caught this pair. Excluding f_1 and f_3 from the model lowers the maximum VIF from 15.74 to 2.49. The new pre-closure flag passes both screens ($\rho = -0.40$ with days, VIF 2.32, Fig. 5c) and is kept.

Mutual information, computed against leave-one-out targets, is reported but not used for selection. The values for f_5 (0.68), f_3 (0.53), and days (0.52) are high relative to their correlations. Nearest-neighbor estimators [8] assume continuous data, and these features have many tied values, so these estimates are interpreted with caution.

VII. Feature-to-target association

Because $R(c)$ is built from the indicators, each indicator is tested against the leave-one-out score R_{-k} that excludes it. The test follows the variable type: Spearman for continuous features, point-biserial r for binary flags, Mann-Whitney for the rare terminated flag, and Kruskal-Wallis for the category. A rule fixed before the tests stated that no feature would be dropped for weak association, since $R(c)$ is a heuristic, not ground truth.

Fig. 6 shows the results. All effects are small ($|\text{effect}| \leq 0.12$) except supplier missing ($r = 0.29$). Category has a small effect ($H = 1{,}300$, $\varepsilon^2 = 0.018$). The terminated flag is inconclusive (rank-biserial -0.12, $p = 0.06$, 86 positives). Value is nearly unrelated to $R(c)$ ($\rho = -0.01$), so $R(c)$ is not a proxy for contract size.

The supplier-missing effect is mechanical (Table II). The indicators differ greatly in scale: f_3 and f_5 are percentile scores with a mean near 0.5, while f_1 averages 0.043 and f_4 averages 0.005. When f_4 is unavailable, it drops out of the average. Contracts with three available indicators average 0.31, against 0.26 with all four. This challenges two Phase 1 assumptions, that available indicators can simply be averaged and that a common $[0, 1]$ range makes them comparable. For comparability, the original definition is kept and the coverage effect is reported as a limitation. Rescaling f_4 to a percentile score is planned as a sensitivity analysis.

TABLE II
Mean $R(c)$ by number of available indicators (training data)

Indicators available	1	2	3	4
Contracts	228	4,592	25,260	42,505
Mean $R(c)$	0.022	0.256	0.312	0.262

VIII. Transformations and the pipeline

The full pipeline chains an indicator builder, a `ColumnTransformer` and an Isolation Forest. Every parameter (clip thresholds, group medians and modes, scaler statistics, encoder categories and supplier identities) is learned from the training period only. Value passes through zero-as-missing, P1/P99 clipping, group imputation, $\log(1 + x)$ and standardization. Days pass through clipping, imputation, the signed log and standardization. Method level, a one-hot category (first level dropped), f_4 and f_5 with missing indicators, and five binary flags complete the 14 inputs. Standardization does not affect the Isolation Forest, which is invariant to per-feature affine transforms. It is needed for VIF and PCA.

Fig. 7 and Table III track each stage. Value skew falls from 137.48 to -0.23. For days, the signed log after clipping reduces kurtosis (10.92 to 4.34) but increases the magnitude of skew (-0.43 to -1.87). The Phase 1 skew rationale therefore does not hold for this feature. The transform is retained as planned, and a clip-only comparison is listed as a sensitivity analysis.

TABLE III
Skewness and kurtosis by transformation stage (training data)

Feature	Stage	Skew	Kurtosis
Value	raw	137.48	20,892.26
	clipped, zeros imputed	6.19	41.32
	$\log(1 + x)$, standardized	-0.23	0.57
Days to signing	raw	-8.33	125.08
	clipped, missing imputed	-0.43	10.92
	signed log, standardized	-1.87	4.34

Table IV verifies the matrix dimensions. No rows are lost in either period and no missing values remain. Synthetic tests confirm that fitted parameters do not change when test data is scored, that each test row is transformed independently of the others, and that unseen methods and categories are handled.

TABLE IV
Pipeline output verification ($N \times P$)

Period	Raw	After indicators	Model input
Train	$72{,}585 \times 23$	$72{,}585 \times 43$	$72{,}585 \times 14$
Test	$36{,}399 \times 23$	$36{,}399 \times 43$	$36{,}399 \times 14$

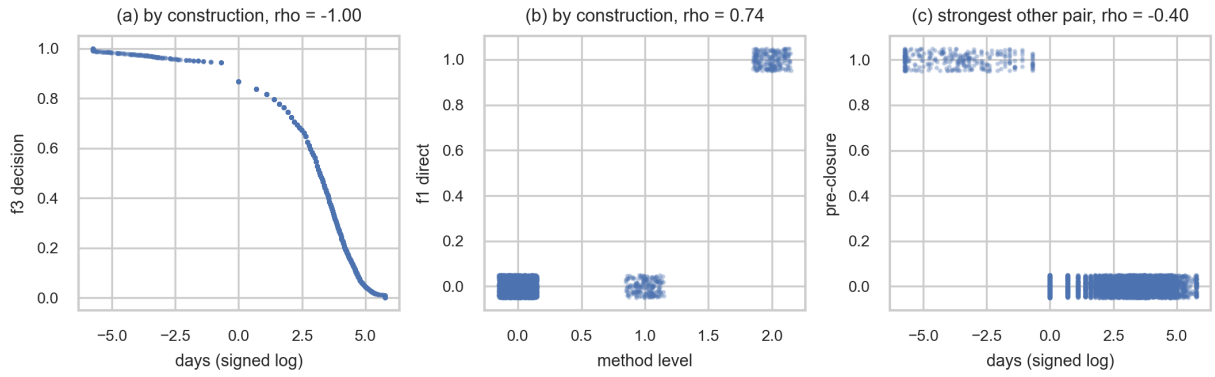


Fig. 5. Collinear pairs: (a) days and f3, by construction; (b) method level and f1, by construction; (c) days and the pre-closure flag, the strongest remaining pair. Samples of 5,000 points with jitter on discrete axes.

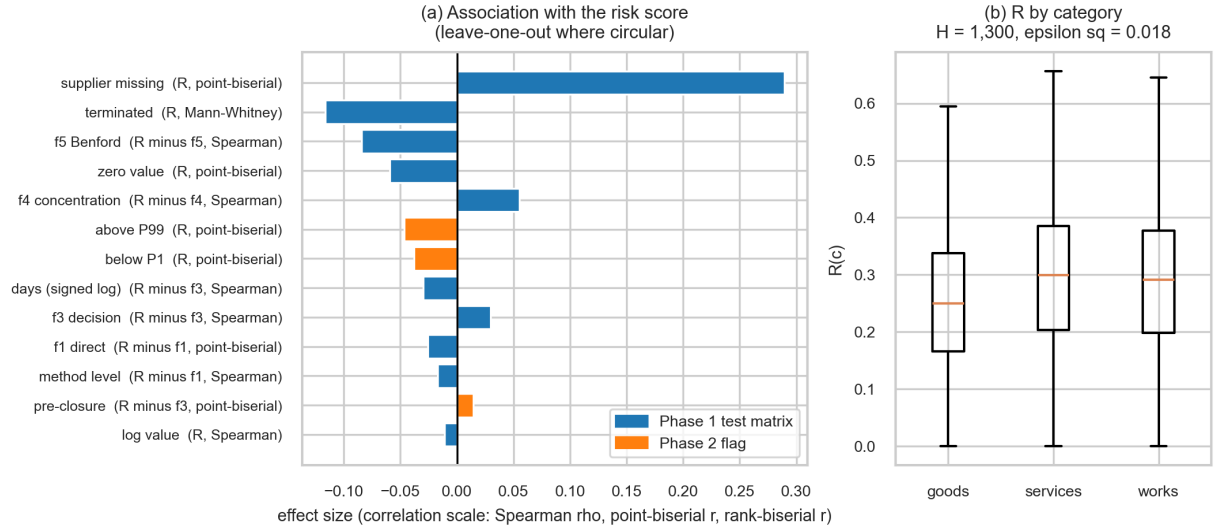


Fig. 6. Feature-to-target association, training data. (a) effect sizes against $R(c)$, leave-one-out where circular; (b) $R(c)$ by procurement category.

IX. Dimensionality

PCA on the 14 standardized model inputs is diagnostic only (Fig. 8). Eleven components are needed for 90% of the variance and twelve for 95%. The first component explains 14.9%, and seven eigenvalues exceed one. The inputs are largely non-redundant, consistent with the maximum VIF of 2.49. The first components mix signing time with value and category (PC1), early signing (PC2), and implausibly small values (PC3). Compressing 14 inputs to 11 would save little and cost interpretability, so the original features are kept.

X. Model Tuning and Validation

Phase 1 tuned the number of trees and the subsample size by rank stability. Stability is the Spearman correlation between the scores of the same validation fold under two random seeds across five time-ordered folds. Contamination is not tuned, because the output is a ranking. Since stability measures consistency rather than sensitivity, synthetic anomaly injection was added. In each validation fold, 2% of contracts were copied after all fitting and perturbed in one of two ways: value multiplied by 50, or signing moved to 200 days before tender close. The copies were scored separately against a top-5% threshold fixed on the clean validation scores, so chance recall is 0.05. Copies never enter any history, fitted parameter or clean ranking.

The selection rule was agreed before the grid was run. It takes the highest mean stability and treats settings within 0.005 of the best as tied. Among tied settings, higher mean injection recall wins. Table V reports all nine settings. The four most stable lie within 0.005, and the rule selects 400 trees with a subsample of 256 (stability 0.9916).

Recall differs sharply by anomaly type. Early signing is detected with a recall of 0.842, about 17 times chance. Value inflation is detected with recall only 0.117, and the mean of the two is 0.480. The model is therefore much less sensitive to this synthetic value perturbation. A plausible mechanism is that a fifty-fold increase usually leaves a typical contract inside the normal range of the whole dataset, and the model has no buyer-

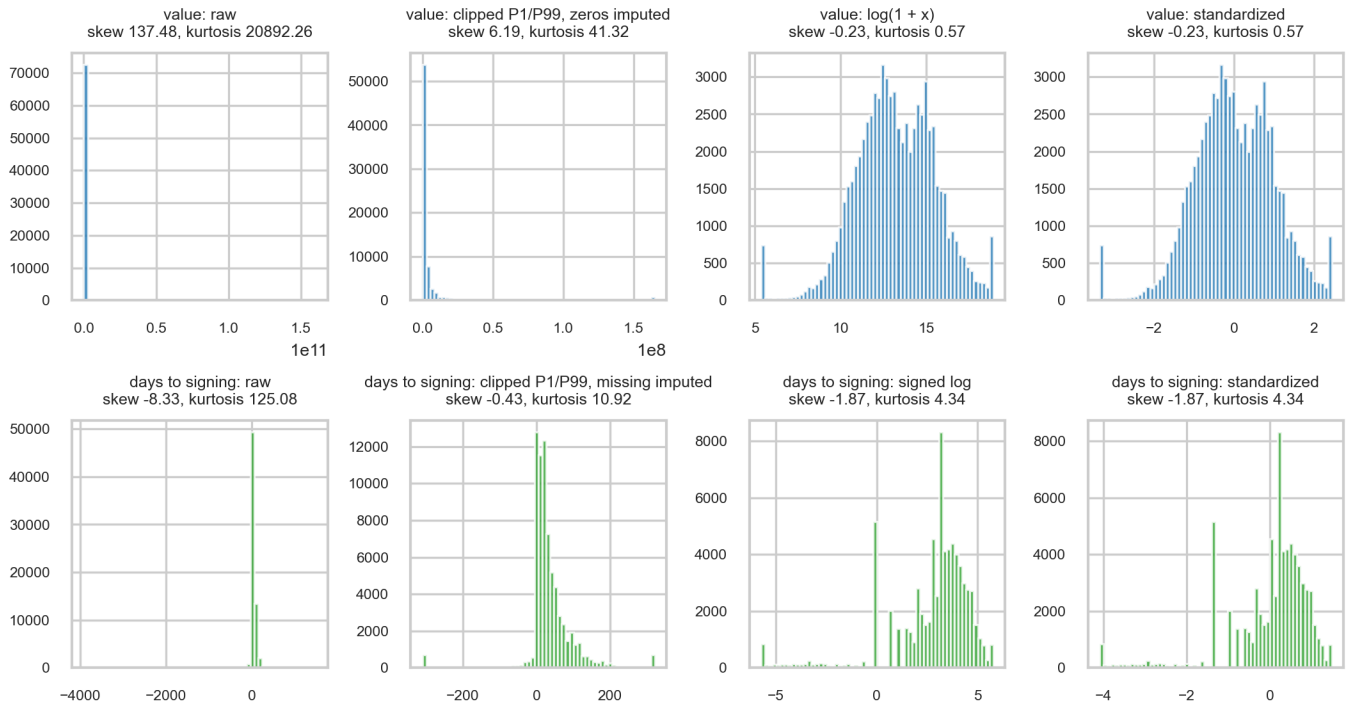


Fig. 7. Transformation stages fitted on training data. Top: contract value. Bottom: days to signing. Spikes at the edges are clipped values; the central spike for days marks imputed medians.

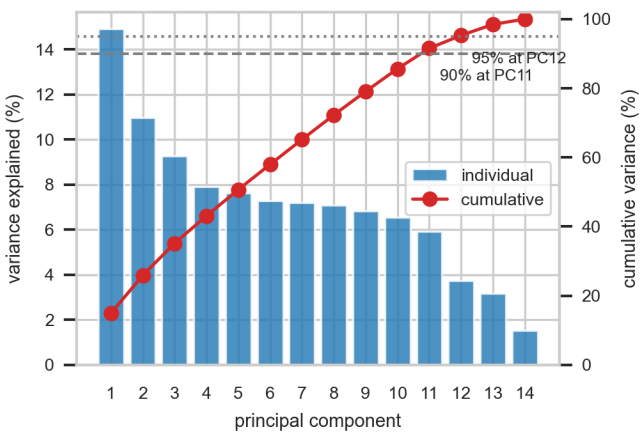


Fig. 8. PCA scree and cumulative variance of the 14 standardized model inputs, training data.

or category-relative value feature. Subsample size trades off the two types: 256 performs better for the date anomaly, while larger samples perform somewhat better for value inflation. Injected anomalies are synthetic, so recall measures sensitivity to these distortions, not to real corruption.

The final model was refitted once on the full training period, and the top-5% threshold was fixed on training scores. In the test period, 3.1% of contracts (1,131) exceed it, against 5.0% in training. This indicates temporal drift whose cause the data cannot attribute: test values are less often extreme, but pre-

TABLE V
ISOLATION FOREST TUNING: MEAN OVER FIVE TIME-ORDERED FOLDS AND TWO SEEDS

Trees	Sample	Stability	Recall value	Recall date	Mean
400	1024	0.9928	0.183	0.732	0.458
400	4096	0.9922	0.195	0.603	0.399
400	**256**	**0.9916**	**0.117**	**0.842**	**0.480**
200	1024	0.9879	0.181	0.717	0.449
200	4096	0.9865	0.195	0.596	0.396
200	256	0.9833	0.116	0.822	0.469
100	4096	0.9771	0.191	0.622	0.407
100	1024	0.9755	0.183	0.732	0.457
100	256	0.9612	0.112	0.801	0.456

closure signing is more common. Model scores agree weakly with $R(c)$ (Spearman 0.145 in training, 0.219 in test). Neither is ground truth, and the two capture different notions of risk.

XI. Decision Log

Table VI compares each Phase 1 plan with the Phase 2 execution and the evidence behind any change.

XII. Limitations and Future Work

No ground-truth corruption labels exist, so neither $R(c)$ nor the anomaly score can be validated as a measure of corruption. $R(c)$ depends on which indicators are available for a contract. The model is much less sensitive to inflated values than to early signing. Entity resolution was validated on pairs by one labeler on small samples, and complete

TABLE VI
STRATEGY VERSUS REALITY: PHASE 1 PLAN AGAINST PHASE 2 EXECUTION

Step	Phase 1 plan	Phase 2 execution	Evidence and reason
Missingness	Tied to procurement method; supplier name and f_2 treated as MNAR	**Changed.** Treated as associated with recording year; MNAR not ruled out	Little's test rejects MCAR ($\chi^2 = 2{,}526.8$, df 16). Cramér's V with method ≤ 0.074 but with year up to 0.903, so missingness is much more strongly associated with the recording period than with the procurement method (Fig. 1)
Award criteria (f_2)	Missing criteria as an MNAR red flag in $R(c)$	**Removed** from $R(c)$ and the model	0.3% missing in 2018 but 98.5 to 100% from 2022: publication ceased, so absence carries no risk signal ($V = 0.903$)
Imputation	Group medians and modes	**Changed.** Value by category (zeros as missing), days and category by method; f_4 and f_5 by median with missing indicators	Year cannot be an imputation group, because test years never occur in training; indicators keep absence visible to the model
Benford (f_5)	At least 100 earlier awards per buyer	**Changed.** Buyer-group fallback, minimum 10	The strict rule covered 62.6% and excluded small buyers; the fallback covers 92.9%
Pre-closure signing	Captured only inside signed-log days	**Added** a binary flag	Keeps a compliance violation explicit: 5.4% of training and 9.0% of test contracts
Supplier identifier	Veto fuzzy merges with `identifier_id`	**Dropped**	552,087 identifiers for 552,087 party rows: not a stable company identifier
Entity resolution	Token sort ≥ 90 within first-token blocks	**Changed.** Strip legal forms, then identical cores or core ≥ 95; no blocking; acronym rule rejected	Only 54% of merges at 90 were correct (184-pair development set). Held-out test: precision 0.98, recall 0.90, against 0.70 and 0.25 for a strict threshold (Table I)
Single bidding (f_6)	Pending validation	**Kept out** of $R(c)$	15 of 50 processes verifiable, and all 15 agree, but below the pre-specified minimum of 30; open tenders only
Value outliers	IQR detection; winsorize at P99, then $\log(1+x)$	**Changed.** Zeros as missing; clip at P1 and P99; $\log(1+x)$; new `below_p1` flag	IQR capping would hit 12.7% (21.2% of works). An upper cap alone does not reduce skew; P1 and P99 take skew from 137.48 to -0.23 (Figs. 3 and 7)
Days outliers	Signed log	**Changed.** Clip at P1 and P99, then signed log	A range of $-3{,}911$ to 1,499 days widens the range from which Isolation Forest split points are drawn; clipping bounds it, and all 3,905 early signings stay negative (Figs. 3 and 7)
Collinearity	Screen $\lvert\rho\rvert \geq 0.85$, VIF > 10	**Kept.** f_1 and f_3 excluded from the model; pre-closure flag retained	Days and f_3: $\rho = -1.00$; f_1 is a function of the method; maximum VIF falls from 15.74 to 2.49 (Figs. 4 and 5)
Target tests	Leave-one-out targets	**Kept.** No feature dropped	$R(c)$ is a heuristic, not ground truth. All effects ≤ 0.12 except supplier missing, $r = 0.29$ (Fig. 6)
Risk score $R(c)$	Mean of available indicators, no rescaling	**Kept** for comparability; limitation reported; rescaling planned	Scales differ (f_4 averages 0.005, f_3 and f_5 about 0.5), so mean $R(c)$ is 0.31 with three indicators against 0.26 with four (Table II)
Scaling	StandardScaler	**Kept**	Needed for VIF and PCA; Isolation Forest is unaffected by scaling
Pipeline	Pipeline and ColumnTransformer, fitted on training data	**Kept**	$72{,}585 \times 23$ to $72{,}585 \times 14$ (test $36{,}399 \times 14$); no rows lost, no missing values (Table IV)
PCA	Diagnostic only	**Kept.** Original features retained	11 of 14 components for 90% and 12 for 95%, so little redundancy to remove (Fig. 8)
Model tuning	Rank stability, five time-ordered folds	**Extended** with anomaly injection; 400 trees, sample 256	The four most stable settings tied within 0.005; recall broke the tie: date 0.842, value 0.117 (Table V)

company renamings cannot be detected from names. Recall is measured only within the candidate pairs. Anomalies outside the training range can be clipped, so the flags carry their signal. f_6 could be verified for only 30% of sampled processes, all of them open tenders. The test period ends in August 2026, so its final year is partial. A ranked per-contract record of indicator values, flags and anomaly scores for the held-out test period is provided in the repository; explanations of individual anomaly scores, including SHAP-based attribution [10] and an interactive auditor review interface, are planned for the modeling and deployment stage.

Planned work includes three sensitivity analyses: f_4 as a percentile score, clip-only days, and a buyer- or category-relative value feature. Future modeling will also compare the current Isolation Forest baseline with alternative anomaly-detection architectures, beginning with Extended Isolation Forest [9], under the same temporal validation and anomaly-injection framework. Further steps include bulk validation of

single bidding with regulator records and an itemized red-flag scorecard for each flagged contract.

XIII. Conclusion

Testing the Phase 1 design against real data changed it in measurable ways. Missingness is strongly associated with the recording period. The supplier identifier does not identify companies. Simple fuzzy matching merges different firms. The lower tail of contract value also matters, and upper-tail winsorization alone is insufficient. The composite score depends on indicator coverage. Each decision is recorded with its evidence. The resulting pipeline is reproducible and designed and tested to prevent temporal leakage. In synthetic validation, it detects the early-signing perturbation far better than the value-inflation perturbation. That gap identifies an important direction for future modeling.

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